

PROJECT REPORT

AdventureWorks Bike Shop

Power BI Sales, Product & Customer Analytics Dashboard

Prepared from: Power BI Dashboard Export (4 report pages)

Subject area: Retail / E-commerce sales analytics (bike & accessories retailer)

Report date: 24 June 2026

Document type: Dashboard Walkthrough & Analysis Report

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Executive Summary

This report documents and analyses a four-page Power BI dashboard built for “AdventureWorks Bike Shop,” a retailer of bikes, accessories, and clothing. The dashboard consolidates sales, product, geographic, and customer data into a single interactive analytics tool intended for management reporting and decision support.

Across the reporting period captured in the dashboard (January 2020 – July 2022), the business recorded \$24.9M in total revenue, \$10.5M in total profit, approximately 25,000 orders, and a 2.2% return rate. The report is organised into four functional pages — Sales Overview, Geography, Product Detail, and Customer Insights — each targeting a different analytical audience (executives, regional/ops teams, product/merchandising teams, and CRM/marketing teams respectively).

Key findings include strong year-over-year revenue growth with seasonal demand peaks, Accessories as the leading order category, Tires and Tubes as the most-ordered product line, Shorts as the most-returned product line, and a customer base concentrated in the “Average” income band and “Professional” occupation segment. Detailed observations, design notes, and improvement recommendations follow in the sections below.

1. Project Overview & Objectives

AdventureWorks is a well-known sample retail dataset (bikes, accessories, clothing and components) commonly used for Business Intelligence training and portfolio projects. This dashboard repurposes that data into a polished, multi-page Power BI report styled with a black side-navigation rail, a teal/cyan brand palette, and card-based KPI tiles.

1.1 Purpose

- Provide a single source of truth for revenue, profit, order volume, and returns.
- Allow drill-down from company-wide KPIs to category, product, country, and customer level.
- Support “what-if” exploration of pricing impact on an individual product’s profit (Product Detail page).
- Profile the customer base by demographics and identify top-value customers.

1.2 Intended Audience

- Executive / leadership team — Sales Overview page.
- Regional and operations managers — Geography page.
- Merchandising / product managers — Product Detail page.
- Marketing / CRM team — Customer Insights page.

1.3 Reporting Period & Scope

The time-series visuals span January 2020 through July 2022 (roughly 2.5 years), giving enough history to evaluate growth trends and seasonality. The data model appears to cover three product categories (Bikes, Accessories, Clothing) sold to customers across at least six countries (United States, Canada, United Kingdom, France, Germany, Australia).

2. Underlying Data Model (Inferred)

While the source .pbix file was not directly inspected, the visuals and fields displayed (e.g., ProductName, Total_orders, Total_revenue, Return_percentage, Full name, income level, occupation) indicate a standard star-schema design typical of the AdventureWorks sample model:

Table	Likely Role	Key Fields Observed
Sales / Orders (fact)	Fact table at order-line grain	Order date, Total_orders, Total_revenue, Total_profit, Total_return_quantity
Product (dimension)	Product attributes & category	ProductName, Category (Accessories / Bikes / Clothing), Return_percentage
Customer (dimension)	Customer demographics	Full name, Income level, Occupation, Country/Region
Geography (dimension)	Country-level location data	Country, Region group (Europe / North America / Pacific)

Date (dimension)	Calendar table for time intelligence	Week, Month, Year — drives all trend charts and year slicers
Targets / Budget	Goal values for gauges	Target order, revenue and profit figures used in Product Detail gauges

The presence of an editable “Price Adjustment” slider alongside an “Adjusted_profit” series on the Product Detail page suggests a what-if parameter (Power BI What-If Parameter feature) used to simulate the profit impact of a price change for the selected product, layered on top of the base DAX measures.

3. Dashboard Architecture

The report consists of four pages, navigated via a persistent black rail on the left edge of every page (a back/home arrow plus icon shortcuts to Geography, Product Detail, and Customer pages):

#	Page Name	Primary Focus	Navigation
1	Sales Overview	Company-wide KPIs, revenue trend, category mix, product league table	Landing page
2	Geography	Order volume by country, regional filter buttons, world map	Accessed via map icon
3	Product Detail	Single-product deep dive with targets, trend, and price what-if	Accessed via drill-through from product table / product icon
4	Customer Insights	Customer base size, demographics, and top-customer leaderboard	Accessed via customer icon

4. Page-by-Page Analysis

4.1 Page 1 — Sales Overview

This is the landing page and functions as the executive summary of the whole report.

KPI Cards (top row)

Metric	Value
Total Revenue	\$24.9M
Total Profit	\$10.5M
Total Orders	25K
Return Rate	2.2%

Trend & Category Visuals

- Weekly Revenue — a line chart from Jan 2020 to Jul 2022 with an overlaid linear trend line and a zoomable range-slider beneath it, showing a clear upward trajectory with weekly volatility and a seasonal late-period spike.
- Orders by Category — a horizontal bar chart ranking Accessories (17.0K orders), Bikes (13.9K), and Clothing (7.0K).

Month-over-Month Cards

Card	Current	Previous Month	Change
Monthly Revenue	\$1.95M	\$1.77M	+10% (positive, teal trend)
Monthly Orders	2,146	2,165	-0.88% (negative, shown in red)
Total Return Quantity	167	169	+1.18% (favourable — fewer relative returns)

Each of these three cards includes a small embedded area sparkline, giving a quick visual sense of the metric's recent trajectory without needing a separate chart.

Product Performance Table

A 10-row table lists individual SKUs with in-cell data bars for Total_orders, alongside Total_revenue and a conditionally shaded Return_percentage column:

ProductName	Total Orders	Total Revenue	Return %
Water Bottle - 30 oz.	3,983	\$39,755	8.48%
Sport-100 Helmet, Red	2,099	\$73,444	3.83%
Sport-100 Helmet, Blue	1,995	\$67,120	3.61%
Sport-100 Helmet, Black	1,940	\$65,270	2.84%
Road Tire Tube	2,173	\$17,265	3.67%
Patch Kit/8 Patches	2,952	\$13,506	5.20%
Mountain Tire Tube	2,846	\$28,333	5.09%
Mountain Bottle Cage	1,896	\$38,062	4.21%
Fender Set - Mountain	1,975	\$87,041	2.95%
AWC Logo Cap	2,062	\$35,882	2.52%

Two callout tiles beneath the table summarise the extremes: Most Ordered Product = “Tires and Tubes” and Most Returned Product = “Shorts” — both at category/sub-category level rather than single SKU, suggesting these are rolled-up groupings distinct from the line-item table above.

4.2 Page 2 — Geography (Total Orders by Country)

A full-page world map (Bing Maps visual) plots bubble markers sized by order volume over six countries: United States and Canada (North America), United Kingdom, France, and Germany (Europe), and Australia (Pacific).

- Three filter buttons — Europe, North America, Pacific — plus a “Select all” reset button sit above the map as a button-slicer bar, letting users isolate a region with one click.
- Bubble size and on-map labels make the United States and Australia visually the largest markers, implying these are the highest-order countries, though exact per-country order counts are not displayed as numeric labels on this page.
- This page is clearly aimed at regional/ops stakeholders who want a quick geographic read on demand distribution rather than granular figures.

4.3 Page 3 — Product Detail (Drill-through page, example: “Water Bottle - 30 oz.”)

This page is a single-product template, almost certainly configured as a Power BI drill-through page so that clicking any row in the Page 1 product table opens this view filtered to that SKU.

Header & Controls

- A large teal title card displays the selected product name.
- A “Price Adjustment” numeric input/slider (currently 0.0) feeds a what-if parameter used to recompute an “Adjusted_profit” figure.
- A checklist (Select all, Total_profit, Total_revenue, Total_return_quantity, Total_orders) lets the user toggle which measures appear in the lower area chart.

Performance vs Target Gauges

Gauge	Actual	Target
Order vs Target	404	438
Revenue vs Target	\$4,067	\$4,292
Profit vs Target	\$2,546	\$2,687

For this example product, actuals are tracking at roughly 92–95% of target across orders, revenue, and profit — close to, but just short of, plan.

Trend Charts

- Total_profit and Adjusted_profit by Month (Jul 2021 – May 2022) — a two-series line chart comparing booked profit against the what-if-adjusted profit; at the current 0.0 price adjustment the two lines fully overlap.
- Total_profit, Total_revenue, Total_return_quantity and Total_orders by Start of Month — a stacked/overlaid area chart letting users compare the shape of all four metrics together, useful for spotting whether returns or orders are the leading driver of revenue swings.

4.4 Page 4 — Customer Insights

Top KPI Cards

Metric	Value
Total Customers	18K
Revenue per Customer	\$1.37K

Demographic Donut Charts

- Orders by Income Level — High: 3K, Average: 12K, Low: 10K — the “Average” income band drives the most order volume.
- Orders by Occupation — Professional: 8K, Skilled Manual: 6K, Management: 4K — “Professional” is the leading occupation segment.

Customer Revenue Trend

A weekly line chart titled “Total Customers by Revenue” (Jan 2020 – Jul 2022) shows revenue rising steadily and accelerating in the most recent period, mirroring the overall company growth seen on the Sales Overview page.

Top 100 Customers Table

A sortable leaderboard (sorted descending by Total_revenue) with a year filter (2020 / 2021 / 2022) above it. The top rows visible are:

Full Name	Total Orders	Total Revenue
Mr. Maurice Shan	6	\$12,408
Mrs. Janet Munoz	6	\$12,015
Mrs. Lisa Cai	7	\$11,330
Mrs. Lacey Zheng	7	\$11,086
Mr. Jordan Turner	7	\$11,022
Mr. Franklin Xu	6	\$10,863
Mr. Larry Munoz	7	\$10,852
Mrs. Kate Anand	4	\$10,437
Mr. Larry Vazquez	4	\$10,395

Total (Top 100)	373	\$8,85,307
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Note: the “Total” row revenue figure (\$8,85,307) is reproduced here exactly as it appears in the source dashboard; the comma placement reflects an Indian numbering display format rather than the standard Western grouping used elsewhere on the page.

Top Customer Callout

A dedicated panel highlights the single highest-revenue customer for the selected year: Mr. Maurice Shan, with 6 orders and \$12K in revenue — echoing the top row of the leaderboard table.

5. Key Business Insights

1. Healthy, sustained growth: Weekly and monthly revenue trend lines both show a consistent upward slope across the 2.5-year window, with monthly revenue up 10% versus the prior month at the time of the snapshot.
2. Orders softening slightly even as revenue grows: Monthly order count fell -0.88% month-over-month while monthly revenue rose +10%, implying an increase in average order value or a shift toward higher-priced items (e.g., bikes over accessories).
3. Accessories drive volume, not necessarily margin: Accessories lead in order count (17.0K), but individual high-return, lower-priced SKUs (e.g., the Water Bottle at 8.48% returns) sit in this category — worth checking whether category-level profit per order is lower than Bikes.
4. Returns are concentrated in a few product lines: “Shorts” is called out as the single most-returned product, and the Water Bottle’s 8.48% return rate is roughly double the next-highest SKU on the table — both are good candidates for a quality or sizing/fit investigation.
5. Geographic concentration: The map highlights the United States and Australia as the most prominent markets by bubble size, with a secondary cluster of UK/France/Germany in Europe and a single Canadian market in North America — a relatively narrow six-country footprint for a brand of this revenue scale.
6. Customer base skews “mainstream”: The largest demographic segments are Average income and Professional occupation, suggesting the core buyer is a working professional with moderate (not high) disposable income — useful for targeting future marketing creative.

7. High customer concentration risk is low: Even the #1 customer by revenue (Mr. Maurice Shan) contributes only \$12,408 against an \$24.9M total revenue base — a long-tail customer base with no single-account dependency.
8. Product-level performance trails target: In the Product Detail example shown, the selected product is running at ~92–95% of its order, revenue, and profit targets, which — if representative — suggests modest underperformance against plan worth monitoring at the category level.

6. Interactivity & UX Features

- Cross-page navigation rail — persistent icon buttons on the left edge of every page for one-click movement between Overview, Geography, Product, and Customer pages.
- Drill-through page — the Product Detail page is built to be entered “in context” from a specific product row, a core Power BI technique for keeping a report uncluttered while still offering depth.
- What-if parameter — the Price Adjustment control on the Product Detail page lets a user simulate a price change and immediately see the effect on an “Adjusted_profit” line, without altering the underlying data.
- Button-style slicers — used for region filtering on the Geography page (Select all / Europe / North America / Pacific) and year filtering on the Customer page (2020 / 2021 / 2022), both friendlier for end users than default dropdown slicers.
- Measure-selector checkboxes — the Product Detail page lets users choose which of four measures populate the area chart, supporting comparative “what matters to me” exploration.
- Range-slider time controls — zoomable date-range sliders sit beneath the Weekly Revenue and Customer Revenue trend charts.
- Sparkline KPI cards — the three Monthly Revenue / Orders / Returns cards each embed a mini area chart for at-a-glance trend context alongside the headline number.

7. Design & Visual Style

- Brand palette — a consistent black / cyan-teal colour scheme is used throughout, anchored by the AdventureWorks gear logo on the landing page.
- Card-based KPI layout — rounded teal tiles for top-line metrics create clear visual hierarchy and are repeated consistently across all four pages.
- Conditional formatting — data bars inside table cells (orders, revenue) and colour-coded return percentages let users scan performance without reading every number.
- Consistent navigation chrome — the same black side rail and back arrow on every page reduces cognitive load when moving between pages.
- Appropriate chart choice — line/area charts for time trends, horizontal bars for category ranking, donuts for compositional demographic splits, and gauges for target attainment — each visual type matches its analytical job.

8. Recommendations & Potential Enhancements

9. Add explicit per-country KPI labels (orders, revenue) on the Geography page — bubble size alone is hard to compare precisely between similarly-sized markets.
10. Surface category-level profit margin (not just order count) on the Sales Overview page to confirm whether Accessories' volume leadership translates into proportional profit.
11. Add a year-over-year or month-over-month % change indicator next to the four top KPI cards on Page 1, similar to the treatment already used on the Monthly Revenue/Orders/Returns cards.
12. Investigate and annotate the root cause of the Water Bottle's elevated 8.48% return rate and the "Shorts" most-returned callout — a tooltip or linked root-cause page would help merchandising teams act on the insight rather than just observe it.
13. Standardise number formatting — the Top 100 Customers total (\$8,85,307) uses a different digit-grouping convention than the rest of the report; aligning all pages to one locale format would avoid confusion.
14. Consider adding a profit-margin-by-product visual to the Product Detail page alongside the existing revenue/profit-vs-target gauges, to fully close the loop between price adjustment, revenue, and margin.

15. Add a customer lifetime value or repeat-purchase-rate metric to the Customer Insights page to complement the existing top-100-by-revenue snapshot.

Conclusion

The AdventureWorks Bike Shop dashboard is a well-structured, four-page Power BI report that successfully layers company-wide KPIs, geographic distribution, product-level deep dives, and customer demographics into a single cohesive analytics product. Its use of drill-through pages, what-if parameters, and button slicers reflects sound Power BI design practice, and its consistent visual branding makes it easy to navigate. The business itself shows healthy overall growth with a few specific areas — elevated return rates on certain SKUs and order counts softening even as revenue rises — that warrant closer follow-up. The recommendations above are intended to extend the report's analytical depth without disrupting its current clean, navigable design.